```
// C++ code
//
const int trig= 6;
const int echo= 5;
const int lr= 13;
const int ly= 12;
const int lg= 11;
const int p= A5;
long duration;
int distance;
void setup()
{
  pinMode(lr,OUTPUT);
  pinMode(ly,OUTPUT);
  pinMode(lg,OUTPUT);
  pinMode(trig,OUTPUT);
  pinMode(p,INPUT);
  pinMode(echo,INPUT);
  Serial.begin(9600);
}
void loop()
{
  digitalWrite(trig,LOW);
  delayMicroseconds(2);
  digitalWrite(trig,HIGH);
  delayMicroseconds(10);
  digitalWrite(trig,LOW);

  duration= pulseIn(echo,HIGH);
  distance= duration*0.034/2;

  int input=analogRead(p);
  int wdis=map(input,0,1023,10,50);
  Serial.print("Warning distance= ");
  Serial.println(wdis);
  Serial.print("Distance= ");
  Serial.println(distance);
  if (distance>wdis && distance<326)
  {
    digitalWrite(lg,HIGH);
    digitalWrite(ly,LOW);
    digitalWrite(lr,LOW);
    Serial.println("The robot is at a safe distance.");
  }
```

```
  else if (distance>(wdis/2) && distance<wdis)
  {
    digitalWrite(lg,LOW);
    digitalWrite(ly,HIGH);
    digitalWrite(lr,LOW);
    Serial.println("The robot is getting close to the obstacle.");
  }
  else if (distance<=(wdis/2))
  {
    digitalWrite(lg,LOW);
    digitalWrite(ly,LOW);
    digitalWrite(lr,HIGH);
    Serial.println("The robot is dangerously close to the obstacle.");
  }
  else
  { digitalWrite(lg,LOW);
    digitalWrite(ly,LOW);
    digitalWrite(lr,LOW);
    Serial.println("The obstacle is outside the range of the ultrasonic sensor.");
  }
}
```